



COMSATS University Islamabad,
Sahiwal Campus

Project Proposal

(SCOPE DOCUMENT)

for

Smart Khata

Version 1.0

By

Muhammad Talha

CIIT/FA23-BSE-041/SWL

Abdul Rehman

CIIT/FA23-BSE-005/SWL

Supervisor

Mr. Syed Ameer Ahmad Gillani

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Project Category:

- A-Desktop Application/Information System ○ B-Web Application/Web Application based Information System ● C-Problem Solving and Artificial Intelligence ○ D-Simulation and Modelling
- E-Smartphone Application ○ F-Smartphone Game ○ G-Networks ○ H-Image Processing
- Other (specify category) _____

Abstract

We built Smart Khata as a smartphone app made specifically for small Pakistani shop owners who currently have no practical digital tool for running their business. The system runs on Flutter for the mobile side, a Python Fast API server on the backend, and MongoDB in the cloud, with AI features layered on top — voice input that accepts Urdu, Punjabi, and English so that even someone who finds typing uncomfortable can use it with no learning curve.

Across Pakistan, millions of kiriyana store owners still rely on paper notebooks and mental calculations because nothing affordable and easy has ever been designed for them. Smart Khata replaces that paper system with digital tools for managing stock, recording sales, tracking customer credit, handling staff attendance and salaries, logging expenses, and suggesting which products to reorder before they run out. Because the app stores everything locally first and syncs to the cloud when internet is available, it works reliably even in areas where the connection comes and goes.

1. Introduction

The backbone of Pakistan's retail economy is the small informal shop kiriyana stores, grain vendors, general merchandise counters and nearly all of them are run the same way: a notebook behind the counter, a mental picture of what is in stock, and informal agreements with customers about what they owe. The problem is not just that this approach is slow. It is that small errors compound over time into real losses stock that runs out because nobody noticed it was getting low, debts that get forgotten, salaries miscounted at month end because attendance was never tracked properly.

Our project tries to fix this by giving shopkeepers a tool that fits how they already communicate. The app is built in Flutter and connects to a Fast API backend, with MongoDB handling all the persistent data. For the AI components we are integrating OpenAI Whisper for voice-to-text conversion, a multilingual BERT model from Hugging Face for understanding what the user is asking, and a Scikit-learn regression model for predicting which products will need to be restocked.

This document defines the full scope of what we are building the problem we are solving, the modules we are developing, the tools we have chosen, how the work is divided between the two of us, and the timeline we are working toward. It is the agreed baseline between the team, our supervisor, and the FYP committee before we write a line of production code.

2. Problem Statement

We visited several kiriyana stores around Sahiwal before starting to write any code, and the pattern was the same everywhere, a battered notebook sitting next to the cash box that serves as the shop's entire financial record. Credit sales go in it, stock levels are guessed from memory, and salaries at month end are whatever the owner can calculate in their head. This approach works until something goes wrong: a page tears, a customer disputes a balance that was written down wrong, or the wrong amount gets paid to a worker because nobody kept track of how many days they showed up.

Business management software does exist but none of it was built for this audience. QuickBooks is designed for people with accounting training. Vyapar is built around Indian GST laws and has no Urdu support. EasePOS charges a monthly subscription that is hard to justify on thin retail margins. KhataBook only handles the credit ledger side and ignores everything else. Not one of these tools allows a shopkeeper to just speak in Urdu and have the system respond which is, for many of these users, the only interaction method they would use consistently.

The real damage from this gap is that shop owners are making their most important decisions what to buy, how much to stock, who to extend credit to, with essentially no data. They cannot tell you about their exact profit from last week, which product turned over fastest this month, or which customers

have owed money for longer than they should. That is the specific problem this project is designed to solve.

3. Problem Solution for Proposed System

The thinking behind Smart Khata is that the software should fit the user, not the other way around. That is why voice input is not an optional extra it is the primary way to use the app. The shopkeeper taps a microphone button and says what they want in Urdu, Punjabi, or English. Something like "chawal ka stock check karo" or "Ali ka hisaab dikhao" is enough. The system handles everything else no menus to dig through, no form to fill out, no keyboard needed.

Behind the voice interface, the audio gets sent to our Fast API backend where Whisper converts it to text and identifies the language on its own. The transcription then goes to a BERT-based intent classifier we are fine-tuning to recognize commands specific to shop management, checking stock levels, creating orders, looking up a customer balance, marking attendance, pulling a report. Once the intent is clear, the right API function runs, the MongoDB database is updated or queried, and a short plain-language answer comes back to the screen.

On the inventory side, the system also does something predictive. We train a Linear Regression model on the previous 30 days of sales data per product and use it to estimate what will be needed in the coming week. Products where predicted demand will exceed current stock show up in a restock list before they run out. This is a simple but genuinely useful feature a paper notebook cannot do this at all

4. Related System Analysis / Literature Review

Before settling on our feature set, we looked at three applications that come closest to what we are building, to understand what is available and where the gaps are.

Vyapar is probably the most complete small-business app in South Asia right now. Inventory, billing, and basic accounting are all there. But it was designed around India's GST tax system, and the interface has no Urdu support. For a Pakistani shopkeeper it is essentially built for someone else and simply happens to be installable on their phone.

KhataBook took a focused approach by digitizing just the credit ledger, the specific thing shopkeepers already understand and doing it cleanly. It even lets users send WhatsApp reminders to customers with outstanding balances. The problem is it stops there entirely. There is no inventory, no reports, no payroll, and no AI. A user who adopts KhataBook still does not know what their stock situation is or whether their shop made money this month.

EasePOS is the most locally relevant of the three. It is a Pakistani product that generates Urdu receipts and is designed for retail. But it is priced and positioned for medium-sized outlets with a proper tech setup. The subscription cost does not work for a small shop on thin margins, there is no offline mode, no employee management, and nothing resembling voice input or predictive features

Table 1: Related System Analysis with Proposed Project Solution

Application Name	Weakness	Proposed Project Solution
Vyapar	No Urdu support, no voice commands, India-centric tax system	Full Urdu interface, multilingual voice commands in Urdu/Punjabi/English
KhataBook	No inventory management, no Urdu script, no AI features	Complete inventory module with AI gives restock recommendations
EasePOS	Paid subscription, no offline mode, no employee management	Free for small shops, full offline sync, built-in HR module

5. Advantages / Benefits of Proposed System

- Shopkeepers who are not comfortable typing can operate the app entirely by speaking in Urdu, Punjabi, or English no keyboard required, no menus to learn.
- The restock engine looks at real sales history to predict which products will run low within the next week, giving the owner a concrete shopping list before a stockout happens.
- All data is written to the phone first and synced to the cloud automatically in the background, so the app works normally even when there is no internet.
- The digital khata tracks every credit sale and payment per customer with a running balance, replacing the paper notebook and eliminating the argument about who owes what.
- Employee accounts can be created with limited permissions so staff can handle orders and mark attendance without seeing financial data or reports.
- Monthly salaries are calculated automatically from attendance records — no manual arithmetic, no end-of-month disputes.
- The app is free to download and use with no subscription, which means it is genuinely accessible to the smallest shops operating on the thinnest margins.

6. Scope

Smart Khata is an Android mobile app for the owner or manager of a single small retail shop. It covers the full daily cycle of running that shop buying from suppliers, tracking inventory, selling to customers, recording what they owe, managing staff, paying salaries, logging expenses, and reviewing financial performance. The interface is in Urdu and English, and Punjabi works through the voice system.

The backend is a Fast API REST API connected to MongoDB Atlas in the cloud. The Flutter app on the phone maintains a local SQLite copy of everything for offline use. Whisper and BERT run as services on the backend and handle all voice processing. The Scikit-learn forecasting model updates as new order data comes in.

We are not building e-commerce, a customer-facing app, multi-branch support, iOS compatibility, tax filing, or payment gateway integration. These are all realistic future additions but sit outside what a two-person student team can deliver in a single academic year.

7. Modules

7.1 Module 1: Inventory and Supplier Management

This is the first module we are building because every other feature depends on having accurate product data. It stores each item with its Urdu and English name, buying and selling price, current stock level, unit of measurement, the supplier it comes from, and the minimum quantity before a low-stock alert fires. When a supplier delivery comes in, the owner logs a purchase order, and the stock numbers update automatically while the cash book records the outflow. Supplier profiles sit alongside product records so the owner can see at any time what they have bought from each one, what they have paid, and what still needs to be settled. Without this foundation the rest of the system has nothing reliable to work with.

7.2 Module 2: Order and Customer Khata Management

This module handles the entire sales workflow from creating a new order to recording customer payments and maintaining running balances. When a customer makes a purchase, the owner or employee creates an order by selecting products and quantities, after which the system automatically calculates the total, deducts the sold quantities from inventory, and records the transaction. If the customer pays on credit, the outstanding balance is tracked in their khata (ledger) profile, which the owner can review at any time. The module supports both full and partial payments and maintains a complete transaction history per customer including sales, payments, and returns. Customer profiles store name, phone number, address, and type (regular customer, farmer, or business buyer). This module digitizes the traditional paper-based khata system that is ubiquitous in Pakistani retail and reduces the risk of unpaid debts going unnoticed.

7.3 Module 3: AI Voice Command Interface

This module is the main reason we built this project. When we talked to shopkeepers during our field visits, several of them told us they had installed apps before but stopped because typing was too slow or the menus were too confusing. The voice interface is our answer to that. The user presses a microphone button and speaks. The audio goes to the backend, Whisper transcribes it and identifies the language, and then the BERT classifier picks the intent check stock, create an order, look up a customer, mark attendance, run a report. The right function runs, the result comes back as a short readable response. No typing. No menus. No learning curve.

7.4 Module 4: Employee HR and Payroll

Even a shop with two or three staff could get into salary disputes if nobody kept track of who came in and when. This module handles that. Each employee is registered with their role and salary structure either a fixed monthly amount or a daily rate. Attendance is marked daily as present, absent, half day,

or on leave. At month end the system calculates net salaries automatically, deducting proportionally for absences in the case of monthly-salaried employees and counting present days for daily workers. The owner marks salaries as paid and the cash book records the outflow. A process that used to take an hour of manual arithmetic now takes a few taps.

7.5 Module 5: Financial Reporting and Smart Recommendations

This module aggregates data from all other modules to provide the shop owner with actionable financial insights. It generates daily, weekly, and monthly reports covering total revenue, cost of goods sold, gross profit, total expenses, and net profit. A dedicated dashboard screen displays summary cards showing key metrics for the current day immediately. The smart restock recommendation feature uses a Scikit-learn Linear Regression model trained on 30 days of order history to predict the demand for each product over the next 7 days, then compares this prediction with current stock levels to generate a prioritized restock list with recommended order quantities. The module also maintains a full cash book showing every cash inflow and outflow with running balance and supports the generation of PDF invoices for completed orders using the Report Lab library.

7.6 Module 6: Offline Sync and Multi-User Access Control

This module ensures the application is reliable and secure in real-world Pakistani shop environments where internet connectivity can be inconsistent. All core data including products, orders, attendance, and expenses are stored locally in a SQLite database on the device using the sqflite Flutter package, allowing every feature to work fully offline. When internet connectivity is detected, the app automatically pushes out of sync records to the MongoDB backend and pulls any changes made from other devices. Conflict resolution follows a defined policy: server data takes precedence for product catalogue updates while client data takes precedence for orders created offline. The access control sub-module allows the shop owner to create employee accounts with restricted permissions, limiting them to creating orders, marking their own attendance, and viewing the product list, while the owner retains full access to financial data, reports, and sensitive configurations.

8. System Limitations / Constraints

- The voice module works best when the user speaks clearly in a reasonably quiet environment. A crowded shop can be noisy, and regional accents vary quite a bit, both can cause Whisper to transcribe incorrectly, which then feeds wrong input to the intent classifier.
- The demand forecasting model learns from real sales data and needs at least a month of history before its predictions become useful. A brand-new shop will get rough guesses in the first few weeks that improve progressively as more data accumulates.
- We are building for Android only. iOS is technically possible with Flutter but getting Apple developer credentials and testing hardware is not realistic for a student project. Android covers most Pakistani smartphone users anyway.
- The system is designed for one shop per account. An owner who runs two separate locations would need two separate installations. Supporting multiple branches under one owner login is a feature that makes sense commercially but is outside the scope of this FYP.

We are working in two-week Scrum sprints under an Agile approach. The main reason we chose Agile over a waterfall plan is that several parts of this project, voice recognition accuracy, intent classification performance, the quality of the restock predictions, cannot be fully specified before we build and test them. A fixed plan written at the start would have been wrong by the second sprint.

9. Software Process Methodology

The project will follow an Agile development methodology, specifically the Scrum framework with two-week sprint cycles. Agile was chosen over waterfall or other traditional methodologies because the requirements of this application involve several AI components whose behaviour may need to be iteratively refined based on testing and feedback. Scrum allows the team to plan each sprint, develop and test a working increment of the software, and then review and adapt the plan for the next sprint based on what was learned.

Each sprint will begin with a planning meeting where tasks are selected from the product backlog and assigned to team members. At the end of each sprint, a review will be conducted with the supervisor to demonstrate the completed increment and gather feedback. This iterative approach will allow AI component performance, such as voice transcription accuracy and demand forecast quality, to be evaluated early and improved progressively rather than discovered as problems only at the end of the project.

10. Tools and Technologies

Table 2: Tools and Technologies for Proposed Project

Tools & Technologies	Version	Rationale
Flutter	3.19	Cross-platform mobile development for Android
Python Fast API	0.110	High-performance async REST API framework
MongoDB Atlas	7.0	Flexible NoSQL cloud database for JSON documents
SQLite (sqflite)	2.3.2	On-device local database for offline mode
OpenAI Whisper	Large-v3	Multilingual speech-to-text transcription
Hugging Face BERT	Bert-base-multilingual	Intent detection from transcribed text
Scikit-learn	1.4	Machine learning for demand forecasting
Pandas / NumPy	2.2 / 1.26	Data processing and numerical computation
JWT + bcrypt	—	Authentication and password security
Report Lab	4.1	PDF invoice generation
Android Studio	Hedgehog	Mobile development IDE
VS Code	1.88	Backend development IDE
Postman	11.0	API testing and documentation

Git / GitHub	2.44	Version control and collaboration
MS Word / PowerPoint	2021	Documentation and presentation

11. Project Stakeholders and Roles

Table 3: Project Stakeholders for Proposed Project

Stakeholder	Role and Responsibilities
Project Sponsor	COMSATS University Islamabad, Sahiwal Campus, provides academic oversight and evaluation framework
Muhammad Talha	Lead developer responsible for Flutter mobile frontend, voice command module, and project documentation
Abdul Rehman	Backend developer responsible for Fast API REST API, MongoDB schemas, ML model training, and API integration
Project Supervisor	Academic mentor providing technical guidance, reviewing deliverables, and approving scope changes
FYP Committee	Evaluation of project deliverables, presentations, and final defence
Small Shopkeepers (End Users)	Primary users of the application; provide feedback during user testing phase

12. Team Members Individual Tasks / Work Division

Table 4: Team Member Work Division for Proposed Project

Student Name	Registration Number	Responsibility / Modules
Muhammad Talha	CIIT/FA23-BSE-041/SWL	Module 3 (Voice AI), Module 6 (Offline Sync & Auth), Flutter UI design, all mobile screens, API integration in Flutter, SQLite offline storage, voice recording and playback, documentation
Abdul Rehman	CIIT/FA23-BSE-005/SWL	Module 1 (Inventory & Suppliers), Module 2 (Orders & Khata), Module 4 (HR & Payroll), Module 5 (Reports & ML), Fast API backend development, MongoDB schema design, ML model training, PDF generation, REST API testing

13. Data Gathering Approach

Before writing any code, we went out and talked to real shopkeepers. Over two weekends we visited five kiriyana stores and two agricultural supply shops in Sahiwal and Faisalabad, watching how owners handle stock, record sales, and deal with customers who owe money. We asked them what goes wrong most often and what they wish they had. Several of them brought up typing unprompted it came up as the main reason they had stopped using a previous app.

We also wrote up a short questionnaire and handed it to fifteen shopkeepers at a local market. The questions covered how they keep records, which languages they read on their phone, whether they use WhatsApp, and whether they would try a free app if one was built for their kind of shop. The responses were consistent: most keep records in notebooks, almost everyone uses WhatsApp daily, and the majority said they would try the app if it worked in Urdu. That confirmed for us that voice input in Urdu was not a secondary feature it is the thing the whole project must be built around.

14. Concepts

Concept 1: Automatic Speech Recognition (ASR)

Automatic Speech Recognition turns spoken audio into written text. We are using OpenAI Whisper, a large transformer model trained on a massive multilingual dataset that includes Urdu. Whisper is useful for us specifically because it detects the language on its own, no setting required, which matters when a shopkeeper might mix Urdu and Punjabi in the same sentence. Working on this component will teach us how to deploy a heavy ML model as a lightweight API service and how to handle audio uploads from a mobile client efficiently.

Concept 2: Natural Language Processing and Intent Detection

Once speech becomes text, we need to figure out what the user wants to do. This is intent classification. We are fine-tuning a multilingual BERT model from Hugging Face on a dataset of labelled shop commands that we are collecting ourselves, examples of how shopkeepers phrase things like checking stock or looking up a balance in Urdu and English. The model learns to identify the intent and pull out the relevant entities: which product, which customer, which date. This is the most data-intensive part of the project and where we expect to spend the most time on evaluation.

Concept 3: Demand Forecasting with Machine Learning

Demand forecasting means using past sales to predict future demand. Our approach is a rolling Linear Regression model trained on the last 30 days of orders per product, used to estimate how much of each item will be needed in the next seven days. If the prediction exceeds current stock, the product goes on the restock list. The technical challenge is in feature engineering deciding which signals from the sales history are predictive and which are just noise from irregular shopping patterns.

Concept 4: Offline-First Mobile Architecture

Offline-first means the phone's local database is the primary store and the cloud is a synchronized backup, not the other way around. We implement this with SQLite on the device through Flutter's sqflite package. Every write goes to SQLite immediately and a background worker handle syncing to MongoDB when there is a connection. Conflict resolution what happens when the same record was changed in two places while offline is one of the more complex design problems in the project and learning to implement it correctly is one of the things we are most looking forward to.

15. Gantt Chart

The following Gantt chart presents the estimated timeline for the development of AI Smart Khata across the two-semester FYP period, spanning from April 2026 to March 2027.

Task	April	May-June	July	August	Sep-Oct	Nov	Dec-Jan	Feb-March
Scope & Requirements	■ ■ ■ ■	■ ■ ■ ■						
Database Schema Design		■ ■ ■ ■	■ ■ ■ ■					
Auth + User Module			■ ■ ■ ■	■ ■ ■ ■				
Inventory + Orders API				■ ■ ■ ■	■ ■ ■ ■			
HR + Payroll + Expenses					■ ■ ■ ■	■ ■ ■ ■		
AI Voice + NLP Module				■ ■ ■ ■	■ ■ ■ ■	■ ■ ■ ■		
ML Restock Engine						■ ■ ■ ■	■ ■ ■ ■	
Flutter UI (All Screens)				■ ■ ■ ■	■ ■ ■ ■	■ ■ ■ ■	■ ■ ■ ■	
Offline Sync Module							■ ■ ■ ■	■ ■ ■ ■
Testing & Bug Fixing						■ ■ ■ ■	■ ■ ■ ■	■ ■ ■ ■
Documentation & Defence							■ ■ ■ ■	■ ■ ■ ■

Figure 15: Gantt Chart, Smart Khata Development Timeline (April 2026 – March 2027)

16. Mockups

The screens below show what the six main areas of AI Smart Khata will look like to the user. Each one corresponds to one of the modules described in this document. The designs are deliberately simple big buttons, clear Urdu labels, as little text on screen as possible because our users are not experienced with smartphone apps and the interface has to be immediately understandable without any training. Login and registration screens are not included as per the scope document guidelines.

16.1 Home Page Mockup

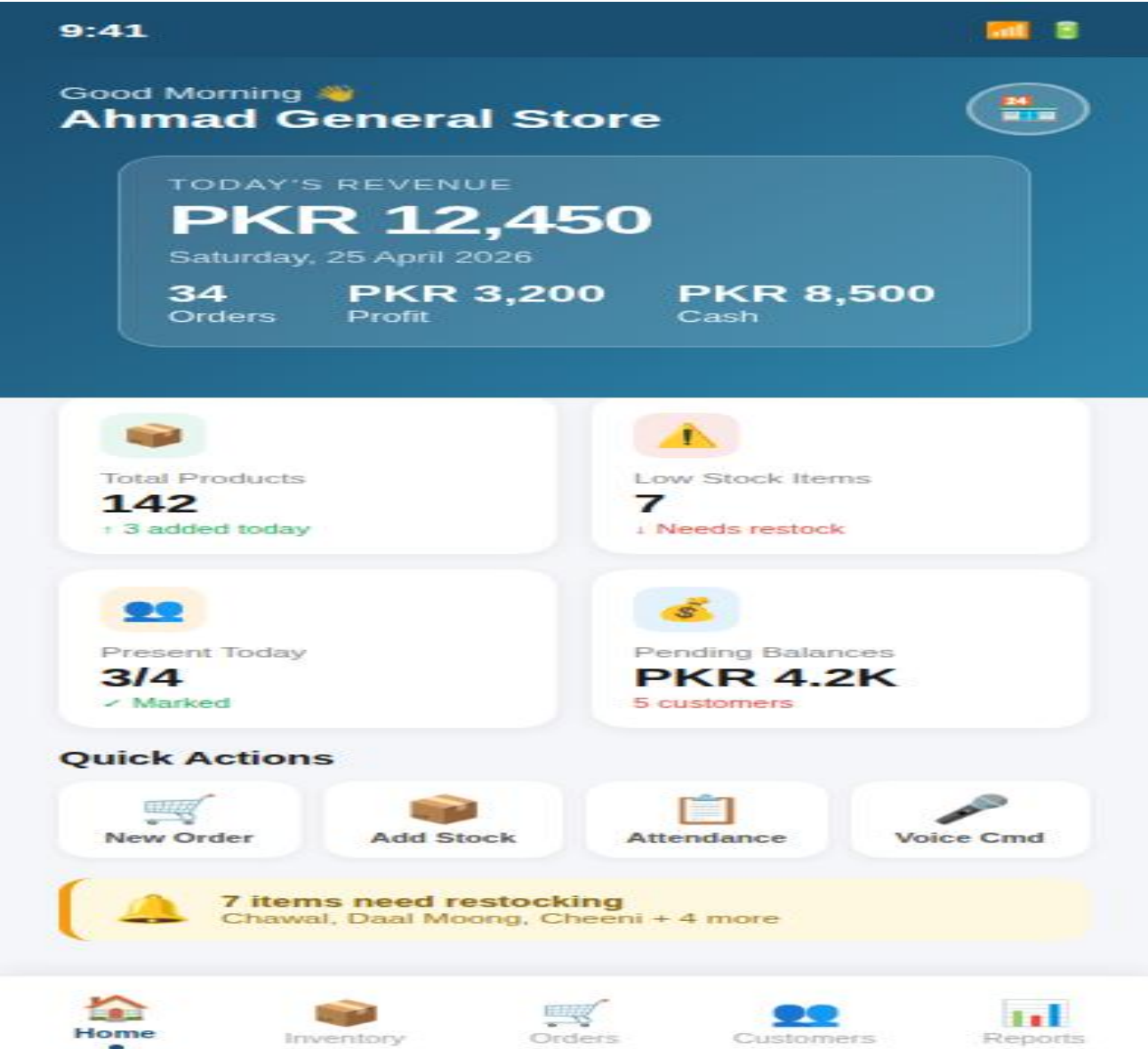


Figure 16.1: Home Page Mockup

16.2 Inventory Management Page Mockup

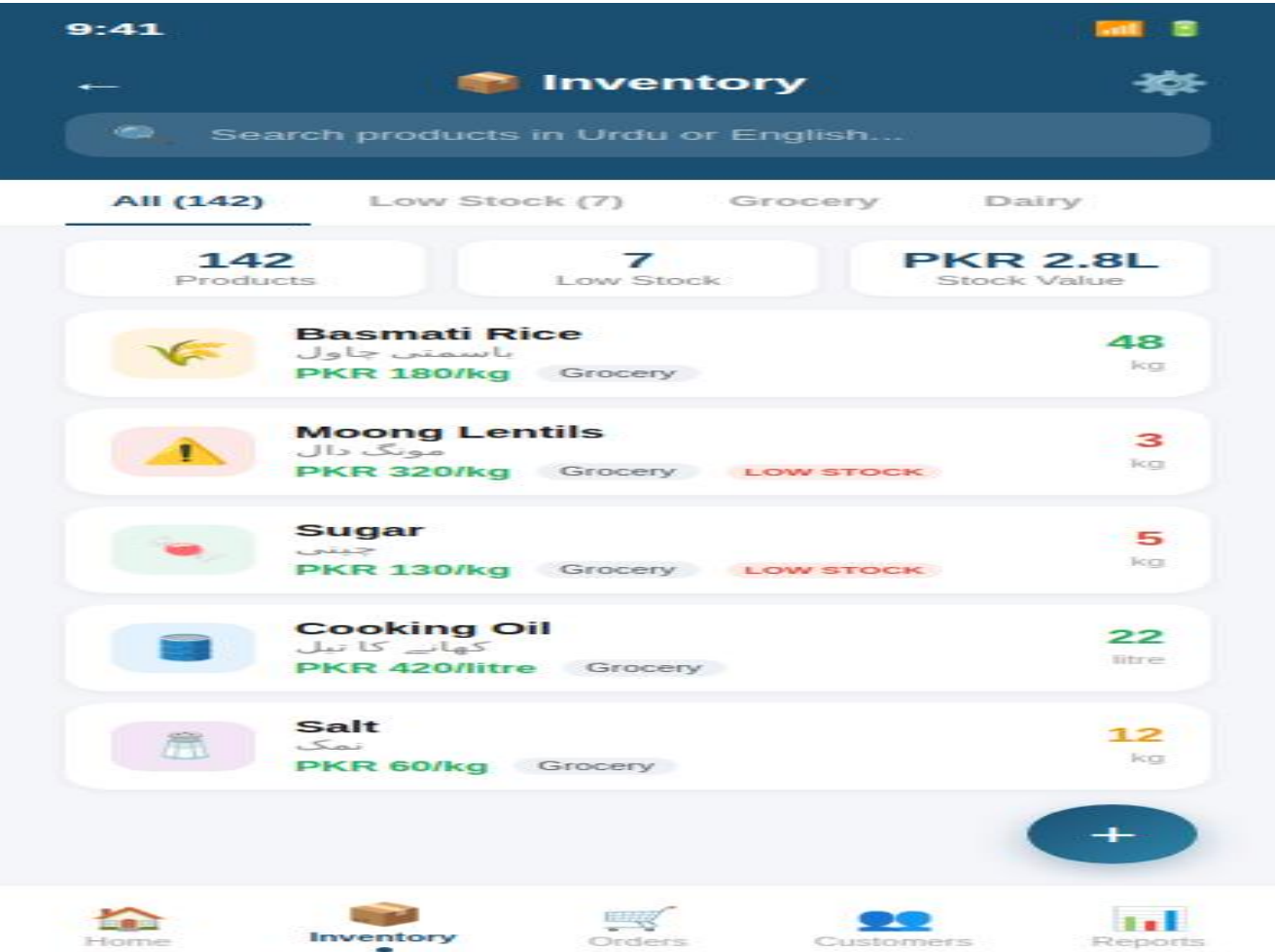


Figure 16.2: Inventory Management Page Mockup

16.3 Orders Page Mockup

9:41

New Order

Order #ORD-2026-0341

Customer name or phone...

Add product by name or barcode...

CART ITEMS (3)

Basmati Rice
PKR 180 × 5 kg

—

5

+

PKR 900

Cooking Oil
PKR 420 × 2 L

—

2

+

PKR 840

Sugar
PKR 130 × 3 kg

—

3

+

PKR 390

Subtotal

PKR 2,130

Discount

— PKR 130

Total

PKR 2,000

PKR 1,500

Balance:

PKR 500

✓ Confirm Order

Figure 16.3: Orders Page Mockup

16.4 Customer Khata Page Mockup

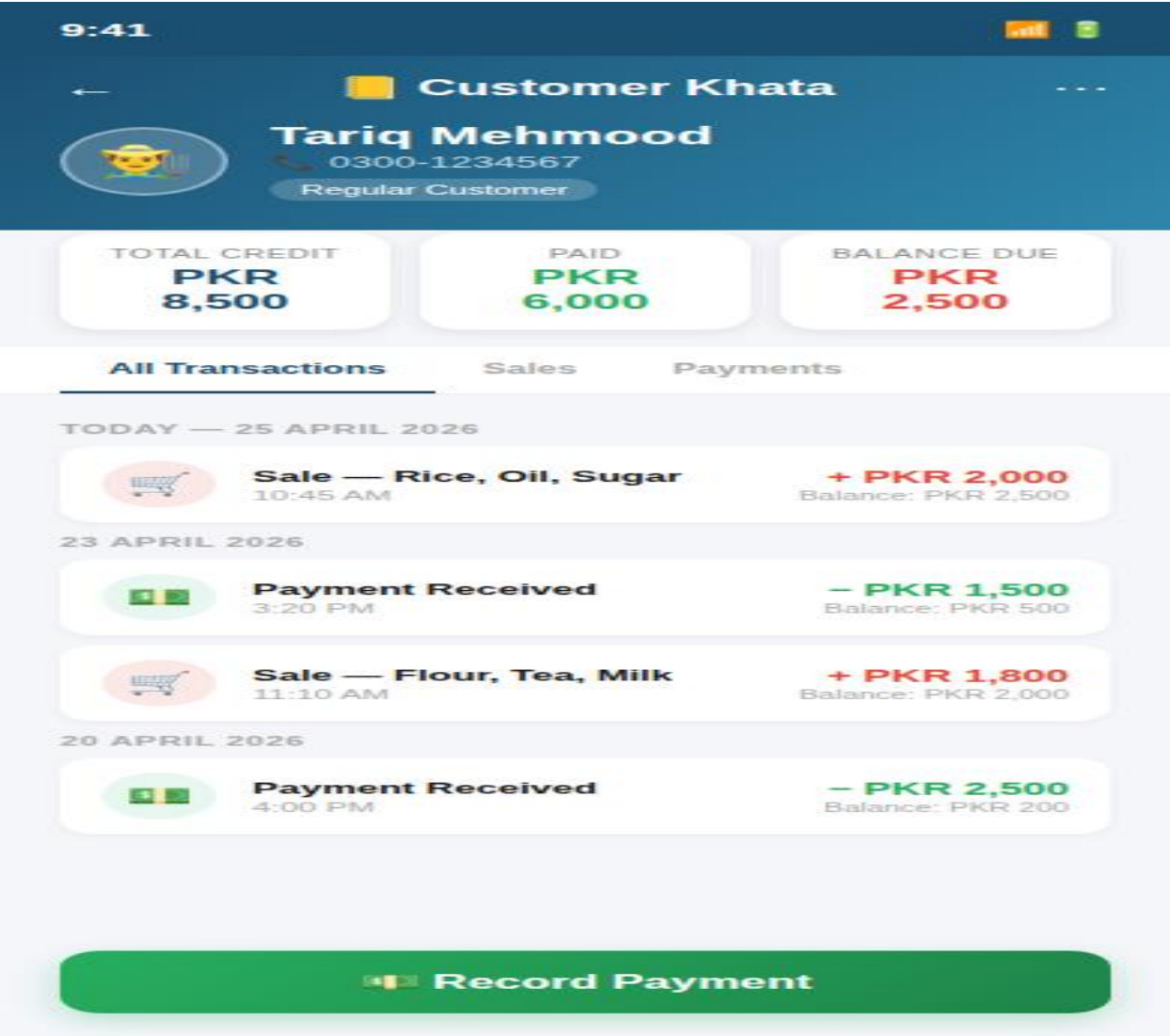
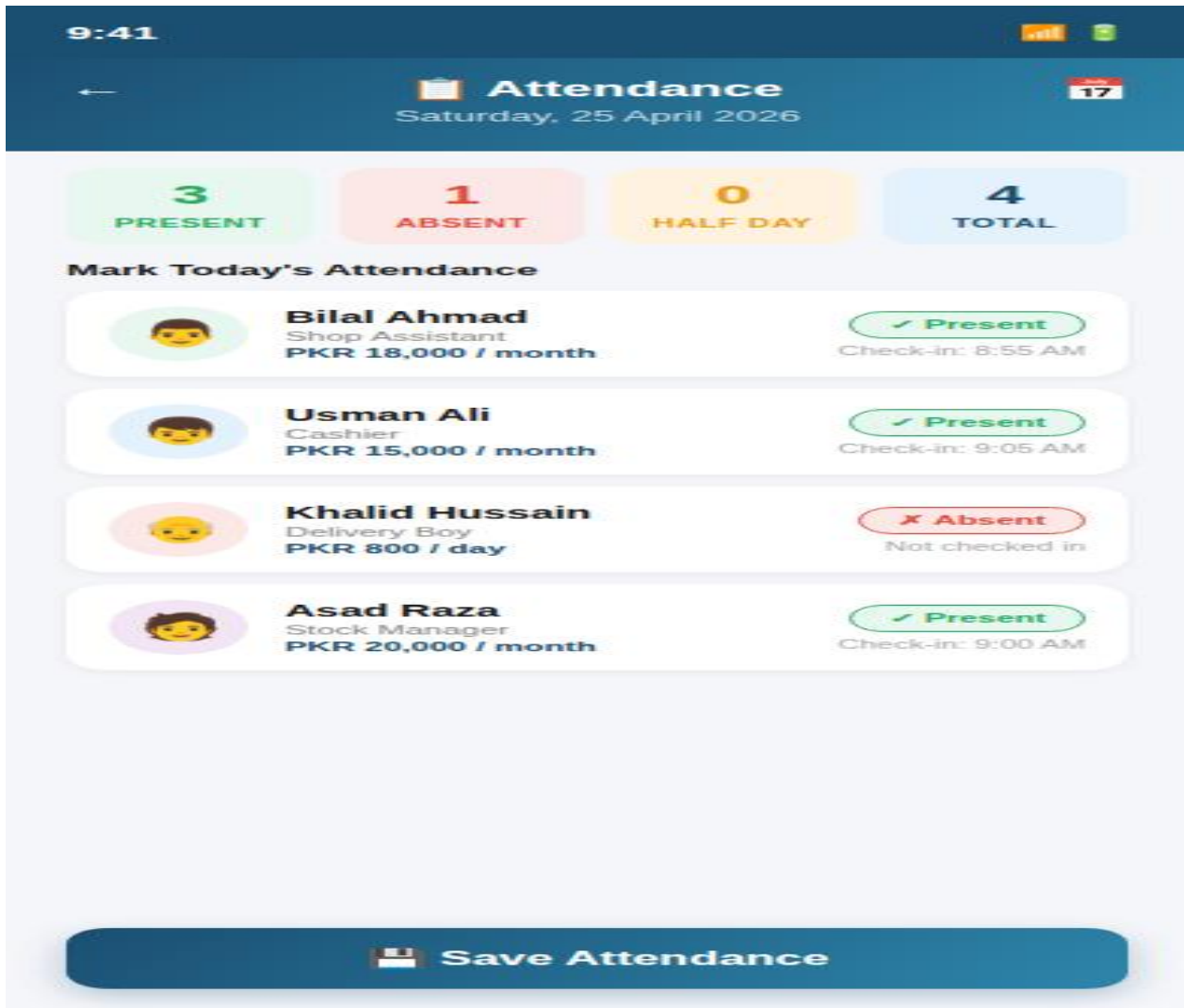


Figure 16.4: Customer Khata Page Mockup

16.5 Attendance Page Mockup



The mockup shows a mobile application interface for tracking attendance. At the top, a dark blue header bar contains the time '9:41' on the left and status icons on the right. Below this, a lighter blue bar features a back arrow, the title 'Attendance' with a calendar icon, and the date 'Saturday, 25 April 2026' with a calendar icon showing '17'. The main content area has a light gray background. It starts with four colored buttons: '3 PRESENT' (green), '1 ABSENT' (red), '0 HALF DAY' (yellow), and '4 TOTAL' (blue). Below these is the section 'Mark Today's Attendance'. This section contains four employee cards. Each card displays an emoji icon, the employee's name, role, and salary. To the right of each card is a status button and a check-in time. The employees are Bilal Ahmad (Shop Assistant, PKR 18,000 / month), Usman Ali (Cashier, PKR 15,000 / month), Khalid Hussain (Delivery Boy, PKR 800 / day), and Asad Raza (Stock Manager, PKR 20,000 / month). At the bottom, a dark blue button with a calendar icon and the text 'Save Attendance' is centered.

9:41

Attendance
Saturday, 25 April 2026

3 PRESENT 1 ABSENT 0 HALF DAY 4 TOTAL

Mark Today's Attendance

 **Bilal Ahmad**
Shop Assistant
PKR 18,000 / month ✓ Present
Check-in: 8:55 AM

 **Usman Ali**
Cashier
PKR 15,000 / month ✓ Present
Check-in: 9:05 AM

 **Khalid Hussain**
Delivery Boy
PKR 800 / day ✗ Absent
Not checked in

 **Asad Raza**
Stock Manager
PKR 20,000 / month ✓ Present
Check-in: 9:00 AM

 Save Attendance

Figure 16.5: Attendance Page Mockup

16.6 AI Voice Command Interface Mockup



Figure 16.6: Voice Command Interface Mockup

17. Conclusion

We started this project because we saw a real and persistent problem that has not been solved for the right audience. Millions of Pakistani shopkeepers run their businesses on paper in 2026 not by choice but because no one has built something that works for them. Smart Khata is our attempt to change that. Whether the app eventually reaches a wide audience or not, building it will leave both of us with real experience across the full stack Flutter, Fast API, MongoDB, speech recognition, NLP, machine learning, and distributed data sync. That is a genuinely broad set of skills built around a problem we care about, grounded in conversations we had with actual shopkeepers before we wrote the first line of code.

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